

open virtual box
New (set any name)
Linux type
next → next → next
Create
last as virtual machine (click)
next
set 50 GB
Create
Settings → Storage → empty (click)
click cd symbol
choose virtual file (any file)
OK
Start
install
☒ check box
Continue
set username : name
password : name@123
Continue
restart now
enter → enter
enter password
popup (ask me later)
Home (search)
terminal

give command edit
(notepad will open)
enter code
save code
Back to terminal
Ctrl + Z
gcc filename.c (compile)
./a.out (run)

```
#include <stdio.h>

int main(){
    int n, i, flag = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    if(n <= 1){
        printf("Not Prime");}
    else {
        for(i = 2; i <= n/2; i++){
            if(n % i == 0){
                flag = 1;
                break;
            }
        }
        if(flag == 0)
            printf("Prime Number");
        else
            printf("Not Prime");
    }
    return 0;
}
```

STEP 1 — Open Google Cloud

Open browser.

Go to:

[Google Cloud Platform](https://cloud.google.com/)

STEP 2 — Login

- Sign in using Gmail account

- Google Cloud page opens
-

STEP 3 — Open Console

Click:

Console

STEP 4 — Create New Project

- Click:

New Project

- Enter project name

Example:

cloud-project

- Click:

Create

STEP 5 — Open Cloud Shell

Click:

Cloud Shell

Terminal opens at bottom.

STEP 6 — Create Project Folder

Type:

`mkdir PythonProject`

Press Enter.

STEP 7 — Open Folder

Type:

`cd PythonProject`

Press Enter.

STEP 8 — Open Editor

Click:

Editor

Editor opens.

STEP 9 — Create main.py File

- Click:

New File

- File name:

main.py

- Press Enter
-

STEP 10 — Type Python Program

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route('/')
def hello():
```

```
    return "Hello World"
```

```
if __name__ == '__main__':
```

```
    app.run(host='127.0.0.1', port=8080, debug=True)
```

Save file:

Ctrl + S

STEP 11 — Create app.yml File

Create new file:

app.yml

Type:

runtime: python
env: flex

entrypoint: gunicorn -b :\$PORT main:app

Save file.

STEP 12 — Create requirements.txt File

Create new file:

requirements.txt

Type:

Flask==2.0.1
gunicorn==20.1.0

Save file.

STEP 13 — Install Required Packages

Back to terminal.

Type:

pip install -r requirements.txt

Press Enter.

Wait until installation completes.

STEP 14 — Run Program

Type:

python main.py

Press Enter.

STEP 15 — Open Preview

Top-right corner:

- Click:

Preview

Then:

Preview on port 8080

OUTPUT

Browser opens.

Output displayed:

Hello World

EXP 4 — Use GAE Launcher to Launch Web Applications

AIM

To create a Google App Engine launcher to launch web applications.

PROCEDURE

Open browser

Go to:

www.cloud.google.com

Login using Gmail account

Click Console

Google Cloud Platform opens

Click Dashboard

Click New Project

Give project name

Example:

cloudHtml

Click Create

Open Cloud Shell

Click Cloud Shell

Terminal opens at bottom

Create Web Project Folder

`mkdir www`

Press Enter.

Then type:

`cd www`

Press Enter.

Create Required Folders

`mkdir css`

`mkdir images`

`mkdir js`

Press Enter after each command.

Create HTML File

`touch index.html`

Press Enter.

Open Editor

Click Editor

Editor opens.

Open index.html

Click:

`index.html`

Type HTML Program

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>

<body>
  <p>Hello World!!!</p>
</body>

</html>
```

Save File

Ctrl + S

Run Program

Back to terminal.

Type:

```
python -m http.server 8080
```

Press Enter.

View Output

Top-right corner:

Click Preview

Click Preview on port 8080

OUTPUT

Browser opens and displays:

Hello World!!!

EXP 5 — Simulate a Cloud Scenario with CloudSim

AIM

To simulate a cloud scenario using CloudSim and run a scheduling algorithm.

PROCEDURE

PART A — Install NetBeans and CloudSim

Download NetBeans IDE

Install NetBeans

Download CloudSim zip file

Extract CloudSim folder

PART B — Create Java Project

Open NetBeans

Click:
File → New Project

Select:
Java

Select:
Java Application

Click Next

Create Project

Enter Project Name

Example:
CloudSimProject

Click Finish

PART C — Add CloudSim Library

Right click Project Name

Click Properties

Click Libraries

Click Add JAR/Folder

Select cloudsim.jar file

Click OK

CloudSim jar file is located inside:

CloudSim/jars

METHOD 1 — IF EXAMPLE PROGRAMS ARE AVAILABLE

Step 1 — Open Example Programs

Open CloudSim folder

Go to:

examples/org/cloudbus/cloudsim

Step 2 — Import Example Files

Drag and drop example files into NetBeans project

Example files:

CloudSimExample1.java

CloudSimExample2.java

CloudSimExample3.java

Step 3 — Run Example Program

Expand:

org.cloudbus.cloudsim.examples

Select:

CloudSimExample1.java

Right click

Run File

OUTPUT

Starting CloudSimExample1...

Initializing CloudSim package

Broker is starting...

Simulation completed

METHOD 2 — IF EXAMPLE PROGRAMS ARE NOT AVAILABLE

Step 1 — Create Java File

Right click Source Packages

New → Java Class

Class Name:
CloudSimExample

Click Finish

Step 2 — Type Program

```
import org.cloudbus.cloudsim.core.CloudSim;  
import java.util.Calendar;
```

```
public class CloudSimExample {  
  
    public static void main(String[] args) {  
  
        System.out.println("Starting CloudSim Simulation");  
  
        try {
```

```
int num_user = 1;
Calendar calendar = Calendar.getInstance();
boolean trace_flag = false;

CloudSim.init(num_user, calendar, trace_flag);

System.out.println("CloudSim initialized successfully");

System.out.println("Simulation completed");

} catch (Exception e) {

    e.printStackTrace();

    System.out.println("Error occurred");
}
}
```

Step 3 — Save Program

Ctrl + S

Step 4 — Run Program

Right click program

Run File

OUTPUT

Starting CloudSim Simulation

CloudSim initialized successfully

Simulation completed

EXP 6 — Transfer Files from One Virtual Machine to Another Virtual Machine

AIM

To transfer files from one virtual machine to another virtual machine using Oracle VirtualBox.

FULL PROCEDURE FROM START TO END

PART 1 — Open VirtualBox

Open Oracle VM VirtualBox

PART 2 — Create First Virtual Machine

Click New

Step 1 — Enter VM Details

Name:

Ubuntu1

Type:

Linux

Version:

Ubuntu (64-bit)

Click Next

Step 2 — Set RAM

Set RAM:

1000 MB

Click Next

Step 3 — Create Hard Disk

Select:

Create virtual hard disk now

Click Create

Step 4 — Select Disk Type

Choose:
VMDK

Click Next

Step 5 — Storage Type

Choose:
Dynamically allocated

Click Next

Step 6 — Set Disk Size

Set:
8 GB

Click Create

PART 3 — Attach Ubuntu ISO

Select Ubuntu1 VM

Click Settings

Click Storage

Click Empty

Click CD icon

Choose Virtual Optical Disk File

Select Ubuntu ISO file

Click OK

PART 4 — Install Ubuntu

Click Start

Ubuntu screen opens.

Step 1 — Install Ubuntu

Click Install Ubuntu

Step 2 — Continue

Click Continue

Step 3 — Installation Type

Select:

Erase disk and install Ubuntu

Click Install Now

Step 4 — Continue

Click Continue

Step 5 — Create User

Enter:

Username

Password

Click Continue

Step 6 — Restart Ubuntu

Click Restart Now

Step 7 — Login

Enter password

Ubuntu opens

PART 5 — Create Second VM

Repeat same steps again.

Create second VM

Name:

Ubuntu2

Install Ubuntu completely.

Now:

- Ubuntu1 VM available
 - Ubuntu2 VM available
-

PART 6 — Change Network Settings

DO THIS FOR BOTH VMs.

Step 1 — Power Off VM

Shutdown Ubuntu VM

Step 2 — Open Network Settings

Select VM

Settings

Network

Step 3 — Change Adapter

Change:

NAT

TO:

Bridged Adapter

Click:

OK

PART 7 — Start Both VMs

Start Ubuntu1

Start Ubuntu2

PART 8 — Find IP Address

In Ubuntu1

Open terminal:

Ctrl + Alt + T

Type:

ifconfig

Press Enter.

Note IP address.

Example:

192.168.1.10

In Ubuntu2

Open terminal.

Type:

ifconfig

Press Enter.

Note second IP address.

Example:

192.168.1.11

PART 9 — Share Folder in Ubuntu2

Step 1 — Create Folder

In Ubuntu2 terminal:

mkdir shared

Press Enter.

Step 2 — Put Files Inside Folder

Copy any file into:

shared

folder.

PART 10 — Connect Ubuntu1 to Ubuntu2

Step 1 — Open File Manager

In Ubuntu1:

Open Files

Step 2 — Connect to Server

Left side:

Click:

Connect to Server

Step 3 — Enter Server Address

Type:

smb://192.168.1.11

(Use Ubuntu2 IP)

Click:

Connect

Step 4 — Enter Credentials

Enter:

Username

Password

Click:

Connect

PART 11 — Access Files

Ubuntu2 shared files appear inside Ubuntu1.

Now:

- Copy files
 - Paste files
 - Transfer files
-

OUTPUT

Files from Ubuntu2 are displayed successfully in Ubuntu1 and transferred successfully.

FULL PROCEDURE FROM START TO END

PART 1 — Open Trystack

Open browser

Go to:

www.trystack.org

Login using account

PART 2 — Create Network

Step 1 — Open Network

Go to:

Network → Networks

Click:

Create Network

Step 2 — Enter Network Name

Network Name:

internal

Click Next

Step 3 — Create Subnet

Network Address:
192.168.1.0/24

IP Version:
IPv4

Click Next

Step 4 — Enter DNS

DNS Name Server:
8.8.8.8

Click Create

PART 3 — Create Instance

Step 1 — Open Instance

Go to:
Compute → Instances

Click:
Launch Instance

Step 2 — Enter Instance Details

Instance Name:
Ubuntu1

Flavor:
m1.medium

Instance Count:
1

Boot Source:
Boot from Image

Image Name:
Ubuntu 14.04 amd64

PART 4 — Configure Access & Security

Step 1 — Import Key Pair

Open:
Access & Security tab

Click:
+ button

Step 2 — Enter Key Pair

Key Pair Name:
mykey

Paste Public SSH Key

Click:
Import Key Pair

Step 3 — Select Security Group

Tick:
default

PART 5 — Configure Networking

Open:
Networking tab

Select created network:
internal

Step 2 — Launch VM

Click:
Launch

PART 6 — Create Router

Step 1 — Open Router

Go to:

Network → Routers

Click:

Create Router

Step 2 — Enter Router Name

Router Name:

router1

Click:

Create Router

Step 3 — Open Router Details

Click router name:

router1

Step 4 — Set Gateway

Click:

Set Gateway

Select:

external

Click OK

Step 5 — Add Interface

Click:

Add Interface

Select subnet

Click:
Add Interface

Step 6 — Open Network Topology

Go to:
Network → Network Topology

PART 7 — Configure Floating IP

Step 1 — Open Instance

Go to:
Compute → Instance

Step 2 — Associate Floating IP

Click:
More → Associate Floating IP

Step 3 — Allocate IP

Click:
+ button

Select:
Pool → external

Click:
Allocate IP

Step 4 — Associate IP

Click:
Associate

Public IP address generated.

Example:
8.21.28.120

PART 8 — Configure Security Rules

Step 1 — Open Security Groups

Go to:

Compute → Access & Security

Open:

Security Groups

Step 2 — Manage Rules

In default row

Click:

Manage Rules

Step 3 — Add ICMP Rule

Click:

Add Rule

Choose:

ALL ICMP

Click Add

Step 4 — Add HTTP Rule

Click:

Add Rule

Choose:

HTTP

Click Add

Step 5 — Add SSH Rule

Click:

Add Rule

Choose:
SSH

Click Add

PART 9 — SSH to Instance

Use floating IP address

SSH into Ubuntu instance

Username:
ubuntu

OUTPUT

Virtual machine launches successfully in Trystack and public IP address is assigned.

EXP 8A — HADOOP INSTALLATION (ONE NODE HADOOP CLUSTER)

AIM

To install Hadoop environment in Ubuntu.

FULL PROCEDURE FROM START TO END

STEP 1 — Open Terminal

Press:

Ctrl + Alt + T

STEP 2 — Go to Downloads Folder

cd Downloads

STEP 3 — Extract JDK

tar zxvf jdk-8u60-linux-x64.gz

STEP 4 — Open JDK Folder

```
cd jdk1.8.0_60/
```

STEP 5 — Check Path

```
pwd
```

Copy displayed path.

Example:

```
/home/cloud/Downloads/jdk1.8.0_60
```

STEP 6 — Edit Profile File

```
sudo nano /etc/profile
```

STEP 7 — Add Java Path

Go to last line and add:

```
JAVA_HOME=/home/cloud/Downloads/jdk1.8.0_60
```

```
PATH=$PATH:$JAVA_HOME/bin
```

```
export PATH JAVA_HOME
```

STEP 8 — Save File

Ctrl + O

Enter

Ctrl + X

STEP 9 — Update Profile

```
source /etc/profile
```

STEP 10 — Install SSH Server

```
sudo apt-get install openssh-server
```

Press:

Y

if asked.

STEP 11 — Generate SSH Key

ssh-keygen

Keep pressing:

Enter

STEP 12 — Copy SSH Key

ssh-copy-id -i localhost

Enter Ubuntu password.

STEP 13 — Go Back

cd ..

STEP 14 — Extract Hadoop

tar zxvf hadoop-2.7.0.tar.gz

STEP 15 — Open Hadoop Folder

cd hadoop-2.7.0/

STEP 16 — Check Hadoop Path

pwd

Example:

/home/cloud/Downloads/hadoop-2.7.0

STEP 17 — Edit Profile Again

```
sudo nano /etc/profile
```

STEP 18 — Add Hadoop Path

Add at end:

```
JAVA_HOME=/home/cloud/Downloads/jdk1.8.0_60
```

```
HADOOP_PREFIX=/home/cloud/Downloads/hadoop-2.7.0
```

```
PATH=$PATH:$JAVA_HOME/bin
```

```
PATH=$PATH:$HADOOP_PREFIX/bin
```

```
export PATH JAVA_HOME HADOOP_PREFIX
```

STEP 19 — Save File

Ctrl + O

Enter

Ctrl + X

STEP 20 — Reload Profile

```
source /etc/profile
```

STEP 21 — Open Hadoop Config Folder

```
cd etc/hadoop
```

STEP 22 — Edit hadoop-env.sh

```
nano hadoop-env.sh
```

Add last line:

```
export JAVA_HOME=/home/cloud/Downloads/jdk1.8.0_60
```

Save and exit.

STEP 23 — Edit core-site.xml

nano core-site.xml

Inside <configuration> add:

```
<property>
<name>fs.defaultFS</name>
<value>hdfs://localhost:9000</value>
</property>
```

STEP 24 — Edit hdfs-site.xml

nano hdfs-site.xml

Inside <configuration> add:

```
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
```

STEP 25 — Create mapred-site.xml

cp mapred-site.xml.template mapred-site.xml

STEP 26 — Edit mapred-site.xml

nano mapred-site.xml

Add:

```
<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>
```

STEP 27 — Edit yarn-site.xml

nano yarn-site.xml

Add:

```
<property>
<name>yarn.nodemanager.aux-services</name>
```

```
<value>mapreduce_shuffle</value>
</property>
```

STEP 28 — Go Back to Hadoop Folder

```
cd ../../
```

STEP 29 — Format Namenode

```
bin/hadoop namenode -format
```

STEP 30 — Start DFS

```
sbin/start-dfs.sh
```

STEP 31 — Check Nodes

```
jps
```

Output:

NameNode

DataNode

SecondaryNameNode

STEP 32 — Start YARN

```
sbin/start-yarn.sh
```

STEP 33 — Check Again

```
jps
```

Output:

NameNode

DataNode

NodeManager

ResourceManager

SecondaryNameNode

STEP 34 — Open Hadoop Browser

Open browser.

Type:

`http://localhost:50070`

STEP 35 — Create Folder in HDFS

`bin/hdfs dfs -mkdir /user1`

STEP 36 — Extract Sample Files

Open new terminal.

Type:

`cd Downloads`

Then:

`tar zxvf mrsampleddata.tar.gz`

STEP 37 — Upload File to HDFS

Go back to Hadoop folder.

`cd hadoop-2.7.0`

Then:

`bin/hdfs dfs -put ../file1.txt /user1`

STEP 38 — Stop Hadoop

`sbin/stop-yarn.sh`

Then:

`sbin/stop-dfs.sh`

OUTPUT

Browser displays Hadoop dashboard successfully and file1.txt appears inside /user1.

EXP 8B — WORD COUNT PROGRAM USING MAP REDUCE TASK

AIM

To execute wordcount program using map and reduce task.

FULL PROCEDURE FROM START TO END

STEP 1 — Open Terminal

Press:

Ctrl + Alt + T

STEP 2 — Go to Hadoop Folder

cd Downloads/hadoop-2.7.0

STEP 3 — Format Namenode

bin/hadoop namenode -format

STEP 4 — Start DFS

sbin/start-dfs.sh

STEP 5 — Check Running Nodes

jps

Output:

NameNode

DataNode

SecondaryNameNode

STEP 6 — Start YARN

sbin/start-yarn.sh

STEP 7 — Check Again

jps

Output:

NameNode

DataNode

NodeManager

ResourceManager

SecondaryNameNode

STEP 8 — Open Hadoop Browser

Open browser.

Type:

http://localhost:50070

STEP 9 — Create Folder in HDFS

Back to terminal.

bin/hdfs dfs -mkdir /user1

STEP 10 — Open New Terminal

Press:

Ctrl + Alt + T

STEP 11 — Go to Downloads

cd Downloads

STEP 12 — Extract Sample Data

tar zxvf mrsampleddata.tar.gz

Output:

file1.txt

file2.txt

file3.txt

file4.txt
file5.txt

STEP 13 — Upload File to HDFS

Go back to Hadoop terminal.

```
bin/hdfs dfs -put ../file1.txt /user1
```

STEP 14 — Check File in Browser

In browser:

- Click:

Utilities

- Click:

Browse the file system

- Open:

/user1

You can see:

file1.txt

STEP 15 — Run GREP Command

In terminal:

```
bin/hadoop jar share/hadoop/mapreduce/hadoop-mapreduce-examples-2.7.0.jar grep  
/user1/* /output '(CSE)'
```

STEP 16 — Run WORDCOUNT Program

```
bin/hadoop jar share/hadoop/mapreduce/hadoop-mapreduce-examples-2.7.0.jar  
wordcount /user1/file1.txt /output1
```

STEP 17 — View GREP Output

```
bin/hdfs dfs -cat /output/*
```

Output:

STEP 18 — View WORDCOUNT Output

```
bin/hdfs dfs -cat /output1/*
```

Output:

```
B.ARCH 9864  
B.TECH(BIO) 9964  
B.TECH(IT) 10000  
BE(AME) 9853  
BE(CIVIL) 10043  
BE(CSE) 9894  
BE(ECE) 10048  
BE(EEE) 9937  
BE(ICE) 9872  
BE(MECH) 9873
```

STEP 19 — Stop Hadoop

```
sbin/stop-dfs.sh
```

Then:

```
sbin/stop-yarn.sh
```

OUTPUT

The wordcount and grep MapReduce programs execute successfully and display output values.

RESULT

Thus the wordcount program using map and reduce task is executed successfully.